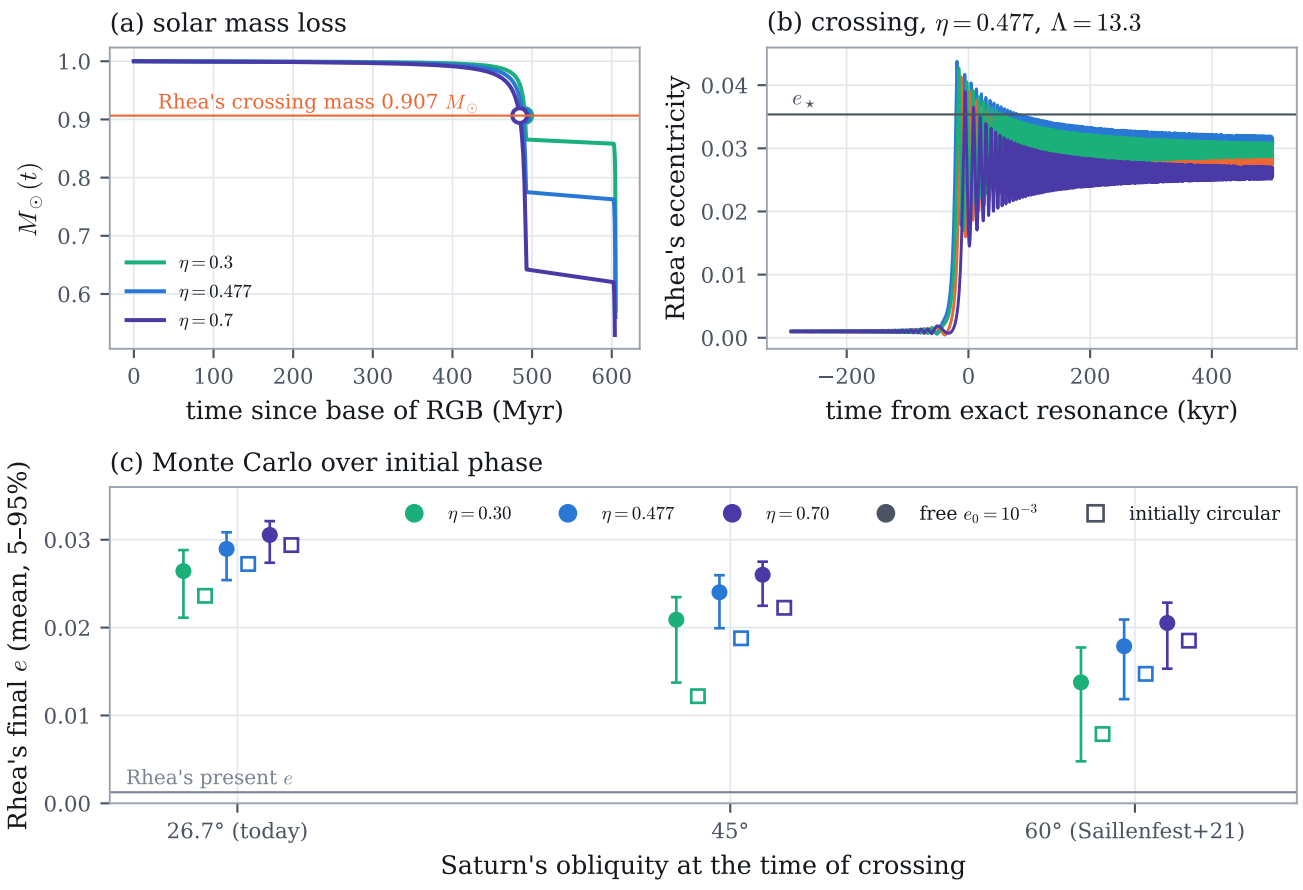

(a) solar mass loss
Rhea's crossing mass 0.907 $M_\odot$
$M_\odot(t)$
$\eta = 0.3$
$\eta = 0.477$
$\eta = 0.7$
time since base of RGB (Myr)
(b) crossing, $\eta = 0.477$, $\Lambda = 13.3$
$e_\star$
Rhea's eccentricity
time from exact resonance (kyr)
(c) Monte Carlo over initial phase
$\eta = 0.30$
$\eta = 0.477$
$\eta = 0.70$
free $e_0 = 10^{-3}$
initially circular
Rhea's present $e$
Rhea's final $e$ (mean, 5–95%)
26.7° (today)
45°
60° (Saillenfest+21)
Saturn's obliquity at the time of crossing